

Amir Reza Vazifeh

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EDUCATION

Princeton University

2023 – 2028

Ph.D., Electrical & Computer Engineering

- GPA 3.9/4.0 | Ph.D. Fellowship in Natural Sciences and Engineering | ECE Graduate Student Award for Excellence in Service (2026)
- **Research:** Computational Imaging & Optics | Computer Vision | AI in Healthcare | Signal Processing
- **Coursework:** Deep Learning Networks | Machine Learning & Pattern Recognition | Statistical Machine Learning | Computer Vision | Imaging the Invisible | Biomedical Imaging | Nonlinear Optics | Digital Signal Processing | Advanced Wireless Systems | Computational Finance in C++

Sharif University of Technology

2018 – 2022

B.S., Electrical Engineering, Minor in Physics

- GPA 18.83/20 | Top 10% of class | Academic Achievement Award (2019–2020) | 12th/134K in National Entrance Exam
- **Research:** Biomedical Signal Processing | AI in Healthcare | Wearable Sensors | Circuit Design | Embedded Systems

EXPERIENCE

Princeton University (Imaging Physics Lab)

2023 – Present

Graduate Researcher — Advisor: Prof. Jason Fleischer; Collaborator: Prof. Felix Heide

Princeton, NJ

- Developing unsupervised video-based image reconstruction in fiber endoscopy using implicit neural representations
- Implementing an algorithm for snapshot hyperspectral imaging with implicit neural representations
- Creating a framework for manifold learning and alignment for personalized multi-lead ECG analysis

Waters Corporation (Industry PhD Partnership)

2024 – Present

Computer Vision and Machine Learning Researcher

Remote, US

- Designing and implementing a deep learning cell counting and viability detection system on brightfield and phase microscopy, robust to focal plane variations and staining strategies
- Driving Waters' next-generation imaging platform; recognized by senior leadership up to SVP level

Institute of Science and Technology Austria

Summer 2023

Research Intern (among top 3% of applicants) — Advisor: Prof. Sandra Siegrt

Vienna, Austria

- Implemented 3D instance segmentation model for microglia cells

EPFL (Integrated Neurotechnologies Lab)

Summer 2022

Research Intern (among top 2% of applicants) — Advisor: Prof. Mahsa Shoaran

Geneva, Switzerland

- Analyzed phase-amplitude cross-frequency coupling in EEG signals

Sharif University of Technology (Biosen Group)

2020 – 2022

Undergraduate Researcher — Advisors: Profs. Mohammad Fakharzadeh & Daryoosh Vashae

Tehran, Iran

- Designed a low-power wearable ECG patch for long-term cardiac monitoring with ML-based arrhythmia detection
- Researched energy harvesting materials and techniques for wearable medical devices

PATENTS

Cell Viability Assessment — U.S. Provisional Patent Application

Assignee: Waters Corporation — Filed: June 2026

Inventors: Christian Zeigler, Amir Reza Vazifeh, Sornanathan Meyyappan, Richard Jeske, Jason W. Fleischer

PUBLICATIONS

- **Under review:** Label-Free Cell Counting and Viability Prediction Using Brightfield Imaging and Deep Learning, **A. R. Vazifeh**, C. Zeigler, S. Meyyappan, R. Jeske, J. W. Fleischer. *Biomedical Optics Express*, 2026
- Neural Fields for Single-Shot Image Reconstruction in Fiber Bundle Imaging Systems, **A. R. Vazifeh**, J. W. Fleischer. *Frontiers in Optics*, 2026
- AI-Enabled Label-Free Cell Viability Assessment from Brightfield Imaging, **A. R. Vazifeh** et al. *Frontiers in Optics*, 2026
- Manifold Alignment for Label-Free Cell Phenotyping in Multimodal Microscopy, **A. R. Vazifeh**, J. W. Fleischer. *Optica Biophotonics Congress — Microscopy, Histopathology and Analytics*, 2026
- Seeing Through Fibers: Unsupervised Image Reconstruction in Fiber Bundle Imaging Systems, **A. R. Vazifeh** et al. *Optics Express*, 2026
- Manifold Learning for Personalized and Label-Free Detection of Cardiac Arrhythmias, **A. R. Vazifeh**, J. W. Fleischer. *Informatics in Medicine Unlocked*, 2026
- An Open-Source Retrospective Analysis of Hypertrophic and Dilated Cardiomyopathy Using Machine Learning and Electrocardiogram Data, A. Altintepe et al. *Diagnostics*, 2026
- Snapshot Hyperspectral Imaging via Compressive Sensing and Implicit Neural Representation, J. Boondicharn, **A. R. Vazifeh**, J. W. Fleischer. *Optica Imaging Congress — Computational Optical Sensing and Imaging*, 2025
- Advancing Personalized Healthcare: Progress in Energy Harvesting Materials of Self-Powered Wearable Devices, P. Bhatnagar et al. *Progress in Materials Science* (Impact Factor: 40), 2023
- Design and Implementation of an Ultralow-Power ECG Patch and Smart Cloud-Based Platform, B. Baraeinejad et al. *IEEE Transactions on Instrumentation and Measurement*, 2022

TEACHING & MENTORSHIP

Princeton University

2024 – 2026

Teaching Assistant & Research Mentor

Princeton, NJ

- TA for Biomedical Imaging (ECE 452) and Machine Learning & Pattern Recognition (ECE 435/535)
- Research mentor, Princeton Laboratory Learning Program (LLP) — mentored high school students in research (Summer 2025)

Sharif University of Technology

2020 – 2022

Teaching Assistant

Tehran, Iran

- TA for Introduction to Machine Learning, Signals & Systems, Computer Architecture, and Applied Electronics in Bioengineering

SKILLS

- **Programming & ML:** Python (PyTorch, TensorFlow, Sklearn, OpenCV, NumPy, SciPy, Pandas), C++
- **Cloud & Data:** AWS (S3, EC2), large-scale data analysis, distributed/parallel computing
- **Hardware:** Circuit analysis & board design (Altium, HSPICE, Proteus), HDL (MIPS Assembly, Verilog)
- **Other:** Git, LaTeX, HTML, CSS

PEER REVIEW

- *IEEE Journal of Biomedical and Health Informatics* (JBHI)
- *The Journal of Supercomputing* (Springer Nature)